

Refinery Unit 4 — Quarterly Maintenance & Inspection Report

Fictional demo document — for RAG ingestion testing only

1. Overview

This report summarizes the quarterly inspection and maintenance activities carried out on the crude distillation unit (CDU-4) and associated rotating equipment. All findings below are fictional and intended solely to test document ingestion pipelines.

2. Equipment Inspected

Equipment	Tag Number	Status
Feed Pump P-401	P-401A	Operational
Heat Exchanger E-410	E-410	Fouling detected
Pressure Relief Valve	PRV-4021	Passed test
Distillation Column	C-401	Operational

3. Findings

- Heat Exchanger E-410 showed a 12% reduction in thermal efficiency compared to baseline, consistent with fouling on the shell side.
- Feed Pump P-401A vibration readings remained within normal operating range (2.1 mm/s RMS).
- Pressure Relief Valve PRV-4021 set pressure verified at 250 psig, within tolerance.

4. Maintenance Recommendations

Schedule chemical cleaning of E-410 shell side within the next 30 days to restore thermal performance. Continue monthly vibration monitoring on P-401A per the mechanical integrity program. No immediate action required for PRV-4021; next test due in 12 months.

5. Safety Compliance Notes

All inspection activities were conducted in accordance with the site's Process Safety Management (PSM) program, including permit-to-work and lockout-tagout procedures. No safety incidents were recorded during this inspection cycle.